

Answer Key — Chemistry Grade 10

Sub-Strand 3.1: Acids and Bases • Lesson 4: Differences Between Acids and Bases Using Commercial Indicators

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
B	B	C	C	A	B	C	C	B	B

USE AFTER: PREDICT PHASE — Prediction task: pre-lab prediction table (neutral row for tap water)

Q1. Before the experiment, Wanjiku predicts what clean tap water will do to litmus paper. Which prediction fits a neutral liquid?

- A. Blue litmus turns red and red litmus stays red
- B. Neither red litmus nor blue litmus changes colour**
- C. Both red litmus and blue litmus turn purple
- D. Red litmus turns blue and blue litmus stays blue

Correct answer: B. A neutral substance such as tap water causes no colour change in either red or blue litmus paper.

USE AFTER: OBSERVE PHASE — Observe, Procedure step 1: litmus strips tested with Solution 1 (lemon juice)

Q2. In the practical, a group adds 2–3 drops of freshly squeezed lemon juice to a strip of blue litmus paper. What colour does the strip turn?

- A. It stays blue
- B. It turns red**
- C. It turns pink
- D. It turns colourless

Correct answer: B. Lemon juice is an acid, and acids turn blue litmus paper red.

USE AFTER: OBSERVE PHASE — Observe, Procedure step 2: phenolphthalein added to each test solution (liquid soap)

Q3. A group adds 2 drops of phenolphthalein to 5 drops of liquid handwashing soap solution. What colour do they see?

- A. Red
- B. Yellow
- C. Pink/magenta**
- D. Colourless

Correct answer: C. Liquid soap solution is a base, and phenolphthalein turns pink/magenta in bases. It stays colourless in acids.

USE AFTER: OBSERVE PHASE — Observe, Procedure step 3: methyl orange added to each test solution (vinegar)

Q4. A group adds 2 drops of methyl orange to white vinegar (siki). What colour should they record?

- A. Yellow
- B. Orange-yellow
- C. Red**
- D. Colourless

Correct answer: C. Vinegar is an acid, and methyl orange turns red in acids. It is yellow in bases and orange-yellow in neutral solutions.

USE AFTER: OBSERVE PHASE — Observe, Launch demonstration and honest-recording reminder

Q5. Before testing, Otieno predicted that lemon juice would turn phenolphthalein pink. In the experiment, the lemon juice and phenolphthalein mixture stays colourless. What should Otieno write in his results table?

- A. Colourless, because he must record exactly what he sees**
- B. Nothing, because the test must have failed
- C. Pale pink, to be halfway between the prediction and the result
- D. Pink, because that is what he predicted

Correct answer: A. Scientists record what they observe, not what they expected. An unexpected result is useful data and must not be changed to fit a prediction.

USE AFTER: EXPLAIN PHASE — Explain: eliciting pattern rules across the ACID row of the consensus table

Q6. A student tests lemon juice with red litmus paper and sees no colour change. She says, "The litmus did not work." Which explanation is correct?

- A. She did not add enough drops, so the test cannot work
- B. Acids turn litmus red, and red litmus is already red, so no change is seen**
- C. Litmus paper only works on bases
- D. Lemon juice is neutral, so nothing happens

Correct answer: B. An acid turns blue litmus red and leaves red litmus red. Red litmus is already red, so no change is visible even though lemon juice is an acid.

USE AFTER: EXPLAIN PHASE — Explain: classification column in the Consensus Results Table

Q7. A solution is tested with four indicators. Red litmus: no change. Blue litmus: no change. Phenolphthalein: colourless. Methyl orange: orange-yellow. How should the solution be classified?

- A. Acid, because phenolphthalein is colourless
- B. Base, because methyl orange is yellow-coloured
- C. Neutral, because all the indicators show the neutral results**
- D. Base, because the litmus papers did not turn red

Correct answer: C. No change in either litmus, colourless phenolphthalein and orange-yellow methyl orange together show a neutral substance, like tap water. Using several indicators avoids wrong conclusions from one result.

USE AFTER: EXPLAIN PHASE — Explain: Pattern Statement for each indicator (phenolphthalein)

Q8. A student adds phenolphthalein to an unknown solution and it stays colourless. What can the student correctly conclude?

- A. The solution is an acid
- B. The solution is neutral
- C. The solution is not a base, but it could be an acid or neutral, so another indicator is needed**
- D. The solution is a base

Correct answer: C. Phenolphthalein is colourless in both acids and neutral solutions and pink only in bases. Colourless means the solution is not a base, but another indicator such as methyl orange is needed to tell acid from neutral.

USE AFTER: DQB PHASE — DQB: 'SOLVED' zone, moving answered questions from Lessons 1–3

Q9. After today's experiment, which question from earlier lessons can the class move to the 'SOLVED' zone on the Driving Question Board?

- A. Will an acid harm our stomach or corrode metal?
- B. Which of these familiar liquids are acids and which are bases?**
- C. Why do acids make some substances fizz?
- D. How can we measure exactly how acidic a liquid is?

Correct answer: B. Testing familiar liquids with litmus, phenolphthalein and methyl orange shows which are acidic, basic or neutral. Measuring how acidic something is and how acids react are still unanswered.

Q10. Revising your model using today's evidence, which set of labels is correct?

- A. Lemon juice: neutral; baking soda solution: base; tap water: acid
- B. Lemon juice: acid; baking soda solution: base; tap water: neutral**
- C. Lemon juice: base; baking soda solution: acid; tap water: neutral
- D. Lemon juice: acid; baking soda solution: neutral; tap water: base

Correct answer: B. Lemon juice is acidic and baking soda solution is basic, based on the indicator results. Tap water is neutral because it causes no colour change in litmus.